

SKILL-7

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SKILL-7:

Aim: Develop a C program to locate a command in the directories listed in the **PATH** environment variable, verify whether it is executable, and execute it in a child process.

The program accepts a command as a command-line argument, reads the PATH variable, searches each PATH directory, checks the discovered path using **stat()** and **access()**, and then uses **fork()**, **execlp()**, and **waitpid()** to execute and synchronize with the child process.

Procedure :

1. Run the program with a command-line argument.
2. Obtain the PATH environment variable using **getenv("PATH")**.
3. Split PATH into individual directories using **strtok()**.
4. Construct a complete path for the requested command and inspect it with **stat()**.
5. Verify executable permission using **access(fullpath, X_OK)**.
6. Create a child using **fork()**; the child executes the command with **execlp()**.
7. The parent waits with **waitpid()** and reports the child's exit status.

Output :

```
Skill > c Skill-7.c
1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <string.h>
4  #include <unistd.h>
5  #include <sys/wait.h>
6  #include <sys/stat.h>
7
8  int main(int argc, char *argv[])
9  {
10     if (argc != 2)
11     {
12         printf("Usage: %s <command>\n", argv[0]);
13         return 1;
14     }
15
16     char *path = getenv("PATH");
17
18     if (path == NULL)
19     {
20         printf("PATH variable not found.\n");
21         return 1;
22     }
23
24     printf("PATH Variable:\n%s\n\n", path);
25
26     char *path_copy = strdup(path);
27     char *dir = strtok(path_copy, ":");
28     int found = 0;
29
30     while (dir != NULL)
31     {
32         char fullpath[1024];
33
34         snprintf(fullpath, sizeof(fullpath), "%s/%s", dir, argv[1]);
35
36         struct stat st;
```

The submitted execution runs `./skill_7 hello80085`. The program prints the PATH environment variable and then reports that the command **hello80085** was not found in any PATH directory.

Code - PATH Search and Process Execution:

```
37
38     if (stat(fullpath, &st) == 0)
39     {
40         if (access(fullpath, X_OK) == 0)
41         {
42             printf("Command found: %s\n", fullpath);
43             found = 1;
44             break;
45         }
46         else
47         {
48             printf("Found but not executable: %s\n", fullpath);
49         }
50     }
51
52     dir = strtok(NULL, ":");
53 }
54
55 free(path_copy);
56
57 if (!found)
58 {
59     printf("Command '%s' not found in PATH.\n", argv[1]);
60     return 1;
61 }
62
63 pid_t pid = fork();
64
65 if (pid < 0)
66 {
67     perror("fork");
68     return 1;
69 }
70
71 if (pid == 0)
```

```

72     {
73         printf("\nChild process started. PID = %d\n", getpid());
74
75         execlp(argv[1], argv[1], (char *)NULL);
76
77         perror("exec failed");
78         exit(1);
79     }
80     else
81     {
82         int status;
83
84         printf("Parent process waiting for child...\n");
85
86         waitpid(pid, &status, 0);
87
88         if (WIFEXITED(status))
89         {
90             printf("Child terminated with exit status: %d\n",
91                   WEXITSTATUS(status));
92         }
93
94         printf("Parent process completed.\n");
95     }
96
97     return 0;
98 }

```

Code (continued):

```
vinay@DESKTOP-0CVM4SE:/mnt/c/Testing/OSSP_Testing/skill$ ./skill_7 hello80085
PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/windows/system32:/mnt/c/windows:/mnt/c/windows/System32/Wbem:/mnt/c/windows/System32/WindowsPowerShell/v1.0:/mnt/c/windows/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/Git/cmd:/mnt/c/Users/venka/AppData/Local/Programs/Python/Python311/Scripts:/mnt/c/Users/venka/AppData/Local/Programs/Python/Python311:/mnt/c/Users/venka/AppData/Local/Programs/Python/Python311/Scripts:/mnt/c/Users/venka/AppData/Local/Programs/Python/Python311:/mnt/c/Users/venka/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/venka/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/venka/AppData/Local/Programs/Antigravity IDE/bin:/mnt/c/Users/venka/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/venka/AppData/Roaming/npm:/mnt/c/Users/venka/.lmstudio/bin:/mnt/c/Users/venka/AppData/Local/Programs/Ollama:/snap/bin

Command 'hello80085' not found in PATH.
vinay@DESKTOP-0CVM4SE:/mnt/c/Testing/OSSP_Testing/skill$ |
```

Functions and Program Operations

Function / Operation	Purpose
getenv()	Obtains the PATH environment variable.
strdup()	Creates a writable copy of the PATH string.
strtok()	Splits PATH into individual directory entries.
snprintf()	Builds the complete command path safely.
stat()	Checks whether the constructed path exists.
access()	Checks whether the command is executable.
fork()	Creates the child process used for command execution.
execlp()	Replaces the child process with the requested command.
waitpid()	Makes the parent wait for the child and obtain its status.

Observation :

- The program requires exactly one command-line argument.
- The PATH variable is displayed and split into individual directories.
- Each candidate command path is checked for existence and executable permission.
- If no executable command is found, the program prints **Command 'hello80085' not found in PATH.**
- When a command is found, the program creates a child process and executes it.
- The parent waits for the child and checks whether it exited normally using **WIFEXITED()** and **WEXITSTATUS()**.

Result :

The SKILL-7 program successfully demonstrates PATH-based command searching, executable permission checking, process creation, command execution, and parent-child synchronization. The submitted test correctly identified **hello80085** as unavailable in PATH.

SKILL-7 Completed Successfully